

ASSIGNMENT 1

Name: Buddhabhushan Sirsat

Roll no.: 99421

1.

Output:

+-----+				
S#	Sname	status	city	
+-----+				
S1	Smith	20	London	
S2	Jones	10	Paris	
S3	Blake	30	Paris	
S4	Clark	20	London	
S5	Adams	30	Athens	
+-----+				

ASSIGNMENT 1

2.

P#	Pname	color	weight	city
P1	Nut	Red	12	London
P2	Bolt	Green	17	Paris
P3	*****	Blue	17	Rome
P4	*****	Red	14	London
P5	Cam	Blue	12	Paris
P6	Cog	Red	19	London

3.

J#	Jname	city
J1	Sorter	Paris
J2	Punch	Rome
J3	Reader	Athens
J4	Console	Athens
J5	Collator	London
J6	Terminal	Oslo
J7	Tape	London

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4.

S#	P#	J#	QTY
S1	P1	J1	200
S1	P1	J4	700
S2	P3	J1	400
S2	P3	J2	200
S2	P3	J3	200
S2	P3	J4	500
S2	P3	J5	600
S2	P3	J6	400
S2	P3	J7	800
S2	P5	J2	100
S3	P3	J1	200
S3	P4	J2	500
S4	P6	J3	300
S4	P6	J7	300
S5	P2	J2	200
S5	P2	J4	100
S5	P5	J5	500
S5	P5	J7	100
S5	P6	J2	200
S5	P1	J4	100
S5	P3	J4	200
S5	P4	J4	800
S5	P5	J4	400
S5	P6	J4	500

ASSIGNMENT 1

Q) Display contents of all tables of database hr

Output:

Tables_in_sandbox_db	
countries	
departments	
employees	
job_history	
jobs	
locations	
regions	

REGION_ID	REGION_NAME
10	Europe
20	Americas
30	Asia
40	Oceania
50	Africa

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AR	Argentina	20
AU	Australia	40
BE	Belgium	10
BR	Brazil	20
CA	Canada	20
CH	Switzerland	10
CN	China	30
DE	Germany	10
DK	Denmark	10
EG	Egypt	50
FR	France	10
GB	United Kingdom of Great Britain and Nort	10
IL	Israel	30
IN	India	30
IT	Italy	10
JP	Japan	30
KW	Kuwait	30
ML	Malaysia	30
MX	Mexico	20
NG	Nigeria	50
NL	Netherlands	10
SG	Singapore	30
US	United States of America	20
ZM	Zambia	50
ZW	Zimbabwe	50

LOCATION_ID	STREET_ADDRESS	POSTAL_CODE	CITY	STATE_PROVINCE	COUNTRY_ID
1000	1297 Via Cola di Rie	00989	Roma	NULL	IT
1100	93091 Calle della Testa	10934	Venice	NULL	IT
1200	2017 Shinjuku-ku	1689	Tokyo	Tokyo Prefecture	JP
1300	9450 Kamiya-cho	6823	Hiroshima	NULL	JP
1400	2014 Jabberwocky Rd	26192	Southlake	Texas	US
1500	2011 Interiors Blvd	99236	South San Francisco	California	US
1600	2007 Zagora St	50090	South Brunswick	New Jersey	US
1700	2004 Charade Rd	98199	Seattle	Washington	US
1800	147 Spadina Ave	MSV 2L7	Toronto	Ontario	CA
1900	6092 Boxwood St	YSW 9T2	Whitehorse	Yukon	CA
2000	40-5-12 Laogianggen	190518	Beijing	NULL	CN
2100	1298 Vileparle (E)	490231	Bombay	Maharashtra	IN
2200	12-98 Victoria Street	2901	Sydney	New South Wales	AU
2300	198 Clementi North	540198	Singapore	NULL	SG
2400	8204 Arthur St	NULL	London	NULL	GB
2500	Magdalen Centre, The Oxford Science Park	OX9 9ZB	Oxford	Oxford	GB
2600	9702 Chester Road	09629850293	Stretford	Manchester	GB
2700	Schwanthalerstr. 7031	80925	Munich	Bavaria	DE
2800	Rua Frei Caneca 1360	01307-002	Sao Paulo	Sao Paulo	BR
2900	20 Rue des Corps-Saints	1730	Geneva	Geneve	CH
3000	Murtenstrasse 921	3095	Bern	BE	CH
3100	Pieter Breughelstraat 837	3029SK	Utrecht	Utrecht	NL
3200	Mariano Escobedo 9991	11932	Mexico City	Distrito Federal	MX

ASSIGNMENT 1

Q) Display contents of all tables of database sales

Output:

| Tables_in_sandbox_db |

| customers |

| orders |

| salespeople |

cnum	cname	city	rating	snum
------	-------	------	--------	------

2001	Hoffman	London	100	1001
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2002	Giovanni	Rome	200	1003
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2003	Liu	San Jose	200	1002
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2004	Grass	Berlin	300	1002
------	-------	--------	-----	------

2006	Clemens	London	100	1001
------	---------	--------	-----	------

2008	Cisneros	San Jose	300	1007
------	----------	----------	-----	------

2007	Pereira	Rome	100	1004
------	---------	------	-----	------

onum	amt	odate	cnum	snum
------	-----	-------	------	------

3001	18.69	1990-10-03	2008	1007
------	-------	------------	------	------

3003	767.19	1990-10-03	2001	1001
------	--------	------------	------	------

3002	1900.10	1990-10-03	2007	1004
------	---------	------------	------	------

3005	5160.45	1990-10-03	2003	1002
------	---------	------------	------	------

3006	1098.16	1990-10-03	2008	1007
------	---------	------------	------	------

3009	1713.23	1990-10-04	2002	1003
------	---------	------------	------	------

3007	75.75	1990-10-04	2004	1002
------	-------	------------	------	------

3008	4723.00	1990-10-04	2006	1001
------	---------	------------	------	------

3010	309.95	1990-10-04	2004	1002
------	--------	------------	------	------

3011	9891.88	1990-10-04	2006	1001
------	---------	------------	------	------

snum	sname	city	comm
------	-------	------	------

1001	Peel	London	0.12
------	------	--------	------

1002	Serres	San Jose	0.13
------	--------	----------	------

1004	Motika	London	0.11
------	--------	--------	------

1007	Rifkin	Barcelona	0.15
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1003	Axelrod	New York	0.10
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Output:				
3200	Mariano Escobedo 9991	11932	Mexico City	Distrito Federal MX
DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID	
10	Administration	200	1700	
20	Marketing	201	1800	
30	Purchasing	114	1700	
40	Human Resources	203	2400	
50	Shipping	121	1500	
60	IT	103	1400	
70	Public Relations	204	2700	
80	Sales	145	2500	
90	Executive	100	1700	
100	Finance	108	1700	
110	Accounting	205	1700	
120	Treasury	NULL	1700	
130	Corporate Tax	NULL	1700	
140	Control And Credit	NULL	1700	
150	Shareholder Services	NULL	1700	
160	Benefits	NULL	1700	
170	Manufacturing	NULL	1700	
180	Construction	NULL	1700	
190	Contracting	NULL	1700	
200	Operations	NULL	1700	
210	IT Support	NULL	1700	
220	NOC	NULL	1700	
230	IT Helpdesk	NULL	1700	
240	Government Sales	NULL	1700	
250	Retail Sales	NULL	1700	
260	Recruiting	NULL	1700	
270	Payroll	NULL	1700	
JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY	
AC_ACCOUNT	Public Accountant	4200	9000	
AC_MGR	Accounting Manager	8200	16000	
AD_ASST	Administration Assistant	3000	6000	
AD_PRES	President	20080	40000	
AD_VP	Administration Vice President	15000	30000	
FI_ACCOUNT	Accountant	4200	9000	
FI_MGR	Finance Manager	8200	16000	
HR_REP	Human Resources Representative	4000	9000	
IT_PROG	Programmer	4000	10000	
MK_MAN	Marketing Manager	9000	15000	
MK_REP	Marketing Representative	4000	9000	
PR_REP	Public Relations Representative	4500	10500	
PU_CLERK	Purchasing Clerk	2500	5500	
PU_MAN	Purchasing Manager	8000	15000	
SA_MAN	Sales Manager	10000	20080	
SA_REP	Sales Representative	6000	12008	
SH_CLERK	Shipping Clerk	2500	5500	
ST_CLERK	Stock Clerk	2008	5000	
ST_MAN	Stock Manager	5500	8500	

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EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	1-515-555-0100	2013-06-17	AD PRES	24000.00	NULL	NULL	90
101	Nene	Yang	NYANG	1-515-555-0101	2015-09-21	AD VP	17000.00	NULL	100	90
102	Lex	Garcia	LGARCIA	1-515-555-0102	2011-01-13	AD VP	17000.00	NULL	100	90
103	Alexander	James	AJAMES	1-590-555-0103	2016-01-03	IT_PROG	9000.00	NULL	102	60
104	Bruce	Miller	BMILLER	1-590-555-0104	2017-05-21	IT_PROG	6000.00	NULL	103	60
105	David	Williams	DWILLIAMS	1-590-555-0105	2015-06-25	IT_PROG	4800.00	NULL	103	60
106	Valli	Jackson	VJACKSON	1-590-555-0106	2016-02-05	IT_PROG	4800.00	NULL	103	60
107	Diana	Nguyen	DNGUYEN	1-590-555-0107	2017-02-07	IT_PROG	4200.00	NULL	103	60
108	Nancy	Gruenberg	NGRUENBERG	1-515-555-0108	2012-08-17	FI_MGR	12000.00	NULL	101	100
109	Daniel	Faviet	DFAVIET	1-515-555-0109	2012-08-16	FI_ACCOUNT	9000.00	NULL	108	100
110	John	Chen	JCHEN	1-515-555-0110	2015-09-28	FI_ACCOUNT	8200.00	NULL	108	100
111	Ismael	Sciarra	ISCIARRA	1-515-555-0111	2015-09-30	FI_ACCOUNT	7700.00	NULL	108	100
112	Jose Manuel	Urman	JMURMAN	1-515-555-0112	2016-03-07	FI_ACCOUNT	7800.00	NULL	108	100
113	Luis	Popp	LPOPP	1-515-555-0113	2017-12-07	FI_ACCOUNT	6900.00	NULL	108	100
114	Den	Li	DLI	1-515-555-0114	2012-12-07	PU_MAN	11000.00	NULL	108	30
115	Alexander	Kho	AKHO	1-515-555-0115	2013-05-18	PU_CLERK	3100.00	NULL	114	30
116	Shelli	Baida	SBAIDA	1-515-555-0116	2015-12-24	PU_CLERK	2900.00	NULL	114	30
117	Sigal	Tobias	STOBIAS	1-515-555-0117	2015-07-24	PU_CLERK	2800.00	NULL	114	30
118	Guy	Himuro	GHIIMURO	1-515-555-0118	2016-11-15	PU_CLERK	2600.00	NULL	114	30
119	Karen	Colmenares	KCOLMENARES	1-515-555-0119	2017-08-10	PU_CLERK	2500.00	NULL	114	30
120	Matthew	Weiss	MWEISS	1-650-555-0120	2014-07-18	ST_MAN	8000.00	NULL	108	50
121	Adam	Fripp	AFRIPP	1-650-555-0121	2015-04-10	ST_MAN	8200.00	NULL	108	50
122	Payan	Kaufling	PKAUFLING	1-650-555-0122	2013-05-01	ST_MAN	7900.00	NULL	108	50
123	Shanta	Vollman	SVOLLMAN	1-650-555-0123	2015-10-10	ST_MAN	6500.00	NULL	108	50
124	Kevin	Mourgos	KMORGOS	1-650-555-0124	2017-11-16	ST_MAN	5800.00	NULL	108	50
125	Julia	Nayer	JNAYER	1-650-555-0125	2015-07-16	ST_CLERK	3200.00	NULL	128	50
126	Irene	Mikkilineni	IMIKKILINENI	1-650-555-0126	2016-09-28	ST_CLERK	2700.00	NULL	128	50
127	James	Landry	JLANDRY	1-650-555-0127	2017-01-14	ST_CLERK	2400.00	NULL	128	50
128	Steven	Markle	SMARKLE	1-650-555-0128	2018-03-08	ST_CLERK	2200.00	NULL	128	50
129	Laura	Bissot	LBISSOT	1-650-555-0129	2015-08-20	ST_CLERK	3300.00	NULL	121	50
130	Mozhe	Atkinson	MATKINSON	1-650-555-0130	2015-10-30	ST_CLERK	2800.00	NULL	121	50
131	James	Marlow	JMARLOW	1-650-555-0131	2015-02-16	ST_CLERK	2500.00	NULL	121	50
132	TJ	Olson	TJOLSON	1-650-555-0132	2017-04-10	ST_CLERK	2100.00	NULL	121	50
133	Jason	Mallin	JMALLIN	1-650-555-0133	2014-06-14	ST_CLERK	3300.00	NULL	122	50
134	Michael	Rogers	MROGERS	1-650-555-0134	2016-08-26	ST_CLERK	2900.00	NULL	122	50
135	Ki	Gee	KGEE	1-650-555-0135	2017-12-12	ST_CLERK	2400.00	NULL	122	50
136	Hazel	Philtanker	HPHILTAN	1-650-555-0136	2018-02-06	ST_CLERK	2200.00	NULL	122	50
137	Renske	Ladwig	RLADWIG	1-650-555-0137	2013-07-14	ST_CLERK	3600.00	NULL	123	50
138	Stephen	Stiles	SSTILES	1-650-555-0138	2015-10-26	ST_CLERK	3200.00	NULL	123	50
139	John	Seo	JSEO	1-650-555-0139	2016-02-12	ST_CLERK	2700.00	NULL	123	50
140	Joshua	Patel	JPADEL	1-650-555-0140	2016-04-06	ST_CLERK	2500.00	NULL	123	50
141	Trenna	Rajs	TRAJS	1-650-555-0141	2013-10-17	ST_CLERK	3500.00	NULL	124	50
142	Curtis	Davies	CDAVIES	1-650-555-0142	2015-01-29	ST_CLERK	3100.00	NULL	124	50
143	Randall	Matos	RMATOS	1-650-555-0143	2016-03-15	ST_CLERK	2600.00	NULL	124	50
144	Peter	Vargas	PVARGAS	1-650-555-0144	2016-07-09	ST_CLERK	2500.00	NULL	124	50
145	John	Singh	JSINGH	44-1632-960000	2014-10-01	SA_MAN	14000.00	0.40	108	80
146	Karen	Partners	KPARTNER	44-1632-960001	2015-01-05	SA_MAN	13500.00	0.30	108	80
147	Alberto	Errazuriz	AERRAZURIZ	44-1632-960002	2015-03-10	SA_MAN	12000.00	0.30	108	80
148	Gerald	Cambrault	GCAMBRAULT	44-1632-960003	2017-10-15	SA_MAN	11000.00	0.30	108	80
149	Eleni	Zlotkey	EZLOTKEY	44-1632-960004	2018-01-29	SA_MAN	10500.00	0.20	108	80
150	Sean	Tucker	STUCKER	44-1632-960005	2015-01-30	SA_REP	10000.00	0.30	145	80
151	David	Bernstein	DBERNSTEIN	44-1632-960006	2015-03-24	SA_REP	9500.00	0.25	145	80
152	Peter	Hall	PHALL	44-1632-960007	2015-08-20	SA_REP	9000.00	0.25	145	80
153	Christopher	Olsen	COlsen	44-1632-960008	2016-03-30	SA_REP	8000.00	0.20	145	80
154	Nanette	Cambrault	NCAMBRAULT	44-1632-960009	2016-12-09	SA_REP	7500.00	0.20	145	80
155	Oliver	Tuvault	OTUVAULT	44-1632-960010	2017-11-23	SA_REP	7000.00	0.15	145	80
156	Janette	King	JKING	44-1632-960011	2014-01-30	SA_REP	10000.00	0.35	146	80
157	Patrick	Sully	PSULLY	44-1632-960012	2014-03-04	SA_REP	9500.00	0.35	146	80
158	Allan	McEwen	AMCEWEN	44-1632-960013	2014-08-01	SA_REP	9000.00	0.35	146	80
159	Lindsey	Smith	LSMITH	44-1632-960014	2015-03-10	SA_REP	8000.00	0.30	146	80
160	Louise	Doran	LDORAN	44-1632-960015	2015-12-15	SA_REP	7500.00	0.30	146	80
161	Sarath	Sewall	SSEWALL	44-1632-960016	2016-11-03	SA_REP	7000.00	0.25	146	80
162	Clara	Vishney	CVISHNEY	44-1632-960017	2015-11-11	SA_REP	10500.00	0.25	147	80
163	Danielle	Greene	DGREENE	44-1632-960018	2017-03-19	SA_REP	9500.00	0.15	147	80
164	Mattea	Marvins	MMARVINS	44-1632-960019	2018-01-24	SA_REP	7200.00	0.10	147	80
165	David	Lee	DLEE	44-1632-960020	2018-02-23	SA_REP	6800.00	0.10	147	80
166	Sundar	Ande	SANDE	44-1632-960021	2018-03-24	SA_REP	6400.00	0.10	147	80
167	Anit	Banda	ABANDA	44-1632-960022	2018-04-21	SA_REP	6200.00	0.10	147	80
168	Lisa	Ozer	LOZER	44-1632-960023	2015-03-11	SA_REP	11500.00	0.25	148	80
169	Harrison	Bloom	HBLOOM	44-1632-960024	2016-03-23	SA_REP	10000.00	0.20	148	80
170	Taylor	Flo	TFLO	44-1632-960025	2016-01-24	SA_REP	9600.00	0.20	148	80
171	William	Smith	WSMITH	44-1632-960026	2017-02-23	SA_REP	7400.00	0.15	148	80
172	Elizabeth	Bates	EBATES	44-1632-960027	2017-03-24	SA_REP	7300.00	0.15	148	80
173	Sundita	Kumar	SKUMAR	44-1632-960028	2018-04-21	SA_REP	6100.00	0.10	148	80
174	Ellen	Abel	EABEL	44-1632-960029	2014-05-11	SA_REP	11000.00	0.30	149	80
175	Alyssa	Hutton	AHUTTON	44-1632-960030	2015-03-19	SA_REP	8800.00	0.25	149	80
176	Jonathan	Taylor	JTAYLOR	44-1632-960031	2016-03-24	SA_REP	8600.00	0.20	149	80
177	Jack	Livingston	JLIVINGSTON	44-1632-960032	2016-04-23	SA_REP	8400.00	0.20	149	80
178	Kimberely	Grant	KGRANT	44-1632-960033	2017-05-24	SA_REP	7000.00	0.15	149	80
179	Charles	Johnson	CJOHNSON	44-1632-960034	2018-01-04	SA_REP	6200.00	0.10	149	80
180	Winston	Taylor	WTAYLOR	1-650-555-0145	2016-01-24	SH_CLERK	3200.00	NULL	128	50
181	Jean	Fleaur	JFLEAUR	1-650-555-0146	2016-02-23	SH_CLERK	3100.00	NULL	128	50
182	Martha	Sullivan	MSULLIVAN	1-650-555-0147	2017-06-21	SH_CLERK	2500.00	NULL	128	50
183	Girard	Geoni	GGEONI	1-650-555-0148	2018-02-03	SH_CLERK	2800.00	NULL	128	50
184	Nandita	Sarchand	NSARCHAND	1-650-555-0149	2014-01-27	SH_CLERK	4200.00	NULL	121	50
185	Alexis	Bull	ABULL	1-650-555-0150	2015-02-20	SH_CLERK	4100.00	NULL	121	50
186	Julia	Dellinger	JDELLINGER	1-650-555-0151	2016-06-24	SH_CLERK	3400.00	NULL	121	50
187	Anthony	Cabrio	ACABRIO	1-650-555-0152	2017-02-07	SH_CLERK	3000.00	NULL	121	50
188	Kelly	Chung	KCHUNG	1-650-555-0153	2015-06-14	SH_CLERK	3800.00	NULL	122	50
189	Jennifer	Dilly	JDILLY	1-650-555-0154	2015-08-13	SH_CLERK	3600.00	NULL	122	50
190	Timothy	Venzl	TVENZL	1-650-555-0155	2016-07-11	SH_CLERK	2900.00	NULL	122	50
191	Randall	Perkins	RPERKINS	1-650-555-0156	2017-12-19	SH_CLERK	2500.00	NULL	122	50
192	Sarah	Bell	SBELL	1-650-555-0157	2014-02-04	SH_CLERK	4000.00	NULL	123	50
193	Britney	Everett	BEVERETT	1-650-555-0158	2015-03-03	SH_CLERK	3900.00	NULL	123	50
194	Samuel	McLeod	SMCLEOD	1-650-555-0159	2016-07-01	SH_CLERK	3200.00	NULL	123	50
195	Vance	Jones	VJONES	1-650-555-0160	2017-03-17	SH_CLERK	2800.00	NULL	123	50
196	Alana	Walsh	AWALSH	1-650-555-0161	2016-04-24	SH_CLERK	3100.00	NULL	124	50
197	Kevin	Feeney	KFEENEY	1-650-555-0162	2016-05-23	SH_CLERK	3000.00	NULL	124	50
198	Donald	OConnell	DOCONNELL	1-650-555-0163	2017-06-21	SH_CLERK	2600.00	NULL	124	50
199	Douglas	Grant	DGRANT	1-650-555-0164	2018-01-13	SH_CLERK	2600.00	NULL	124	50
200	Jennifer	Whalen	JWHALEN	1-515-555-0165	2013-09-17	AD_ASST	4400.00	NULL	101	100
201	Michael	Martinez	MMARTINEZ	1-515-555-0166	2014-02-17	NK_MAN	13000.00	NULL	108	20
202	Pat	Davis	PDAVIS	1-603-555-0167	2015-08-17	NK_REP	6000.00	NULL	201	20
203	Susan	Jacobs	SJACOBS	1-515-555-0168	2012-06-07	HR_REP	6500.00	NULL	101	40
204	Hermann	Brown	HBROWN	1-515-555-0169	2012-06-07	PR_REP	10000.00	NULL	101	70
205	Shelley	Higgins	SHIGGINS	1-515-555-0170	2012-06-07	AC_MGR	12000.00	NULL	101	110

ASSIGNMENT 1

EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
101	2007-09-21	2011-10-27	AC_ACCOUNT	110
101	2011-10-28	2015-03-15	AC_MGR	110
102	2011-01-13	2016-07-24	IT_PROG	60
114	2016-03-24	2017-12-31	ST_CLERK	50
122	2017-01-01	2017-12-31	ST_CLERK	50
176	2016-03-24	2016-12-31	SA_REP	80
176	2017-01-01	2017-12-31	SA_MAN	80
200	2005-09-17	2011-06-17	AD_ASST	90
200	2012-07-01	2016-12-31	AC_ACCOUNT	90
201	2014-02-17	2017-12-19	MK_REP	20

Dark Wiki

ASSIGNMENT 1

Answers

Q1. How many orders data is present in the database?

Answer: There are 10 order records in the orders table (onum 3001, 3002, 3003, 3005, 3006, 3007, 3008, 3009, 3010, 3011).

Q2. How many customers are present in the database?

Answer: There are 7 customers in the customers table: Hoffman, Giovanni, Liu, Grass, Clemens, Cisneros, and Pereira.

Q3. How many salespeople are present in the database?

Answer: There are 5 salespeople in the salespeople table: Peel, Serres, Motika, Rifkin, and Axelrod.

Q4. On which date order with highest amount is placed?

Answer: The highest order amount is 9891.88 (order 3011), placed on 1990-10-04.

Q5. For which order salesman got maximum commission?

Answer: Commission earned = order amount $\times$ salesperson's comm rate. Computing this for every order, the maximum commission is earned on order 3011 (amount 9891.88, salesman Peel, snum 1001, comm rate 0.12), giving a commission of 1187.03 — the highest of all orders.

Next highest for comparison: order 3005 (Serres, comm 0.13 on 5160.45 $\rightarrow$ 670.86) and order 3008 (Peel, comm 0.12 on 4723.00 $\rightarrow$ 566.76).

Q6. Which salesmen are handling more than two customers?

Answer: None. Counting customers per salesman (via the snum column in customers): Peel (1001) $\rightarrow$ 2 customers (Hoffman, Clemens), Serres (1002) $\rightarrow$ 2 customers (Liu, Grass), Axelrod (1003) $\rightarrow$ 1 customer (Giovanni), Motika (1004) $\rightarrow$ 1 customer (Pereira), Rifkin (1007) $\rightarrow$ 1 customer (Cisneros). No salesman handles more than two customers; Peel and Serres each handle the maximum of two.

Q7. Which city have single customer?

ASSIGNMENT 1

Answer: Berlin has a single customer (Grass). London, Rome, and San Jose each have two customers, so Berlin is the only city with just one.

Q8. Which city has multiple salespeople?

Answer: London has multiple salespeople — Peel and Motika are both based in London. San Jose, Barcelona, and New York each have only one salesperson.

Q9. What is name of customer and salesman of the maximum amount order?

Answer: The maximum amount order is order 3011 (amount 9891.88), for customer Clemens (cnum 2006), handled by salesman Peel (snum 1001).

Q10. Study relationship between the tables in sales database.

Answer: The sales database has three entities — customers, orders, and salespeople — connected as follows:

- salespeople → customers: One-to-Many. Each salesperson (snum) can serve many customers, but each customer is assigned to exactly one salesperson (via the snum foreign key in the customers table).
- customers → orders: One-to-Many. Each customer (cnum) can place many orders, but each order belongs to exactly one customer (via the cnum foreign key in the orders table).
- salespeople → orders: One-to-Many. Each salesperson (snum) can be linked to many orders, but each order is handled by exactly one salesperson (via the snum foreign key in the orders table). This is a redundant/derivable link, since the salesperson for an order can also be inferred through the order's customer.

Overall, orders is the 'many' side connected to both customers and salespeople, while customers itself sits in a one-to-many relationship with salespeople — matching the ER diagram given in the assignment (customers *–1 salespeople, customers 1–* orders, orders *–1 salespeople).